Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 08

Name: Vikram Singh Parihar

Roll no. 66 Sec :B

CUID :24250097

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    confusion_matrix, accuracy_score, precision_score,
    recall_score, f1_score, ConfusionMatrixDisplay
)

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

# Convert TotalCharges to numeric and handle missing values
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df = df.dropna()

# Encode categorical variables
X = df.drop(columns=["customerID", "Churn"])
X = pd.get_dummies(X, drop_first=True)
y = LabelEncoder().fit_transform(df["Churn"])

X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.20, random_state=42, stratify=y
)

model = DecisionTreeClassifier(
    criterion="gini", max_depth=5, random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", recall_score(y_test, y_pred))
print("F1 Score :", f1_score(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

ConfusionMatrixDisplay(cm).plot()
plt.title("Confusion Matrix")
plt.show()

plt.figure(figsize=(20, 10))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
    filled=True,
```

```
    max_depth=3
)
plt.title("Decision Tree")
plt.show()

importance = pd.Series(model.feature_importances_, index=X.columns)
print("\nTop influential features:")
print(importance.sort_values(ascending=False).head(10))

# Questions and Answers
# Q1. What is classification, and how does it differ from regression?
# Classification predicts discrete class labels, while regression predicts continuous numerical values.
# Q2. Explain the working principle of the Decision Tree algorithm.
# A Decision Tree recursively splits the data using feature-based rules chosen to make the resulting groups more homogeneous with respect to the target class. Splitting continues until stopping criteria are reached, and a leaf provides the predicted class.
# Q3. Differentiate between Gini Index and Entropy as splitting criteria.
# Both measure impurity. Gini Index is based on the probability of misclassification, while Entropy measures information uncertainty. Lower impurity indicates a purer split.
# Q4. What is a Confusion Matrix? Explain its components.
# A confusion matrix summarizes classification results using True Positives, True Negatives, False Positives and False Negatives.
# Q5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?
# Accuracy is the proportion of all correct predictions. Precision is the proportion of predicted positives that are correct. Recall is the proportion of actual positives detected. F1-Score is the harmonic mean of precision and recall. Together they provide a more complete view than accuracy alone.
# Q6. What is overfitting in a Decision Tree? How can it be reduced?
```

# Overfitting occurs when the tree becomes too complex and models noise in the training data. It can be reduced using max_depth, min_samples_split, min_samples_leaf, pruning, and validation.

# Q7. Why is customer churn prediction important for businesses?

# Early identification of customers likely to leave enables targeted retention actions, improves customer satisfaction and can reduce revenue loss.

# Q8. What are the advantages and limitations of the Decision Tree algorithm?

# Advantages include interpretability, simple rule-based decisions, little need for feature scaling, and support for nonlinear relationships. Limitations include overfitting, instability to small data changes, and potentially poorer generalization than ensembles.

# Q9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

# Credit-risk assessment, medical diagnosis, and fraud detection are three examples.

# Q10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

# Organizations can identify high-risk customers, understand the features associated with churn, prioritize retention campaigns, personalize offers and measure which interventions reduce churn, thereby protecting revenue.

OUTPUT

PS D:\PROGRAMS\Career advancement lab\lab sheet 8> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 8\Experiment_8_code.py"

Accuracy : 0.715

Precision: 0.6535433070866141

Recall : 0.8645833333333334

F1 Score : 0.7443946188340808

Confusion Matrix:

[[60 44]

[13 83]]

Top influential features:

Contract_Two year 0.279331

Contract_One year 0.271344

MonthlyCharges 0.135814

TotalCharges 0.128923

InternetService_Fiber optic 0.098994

tenure 0.061140

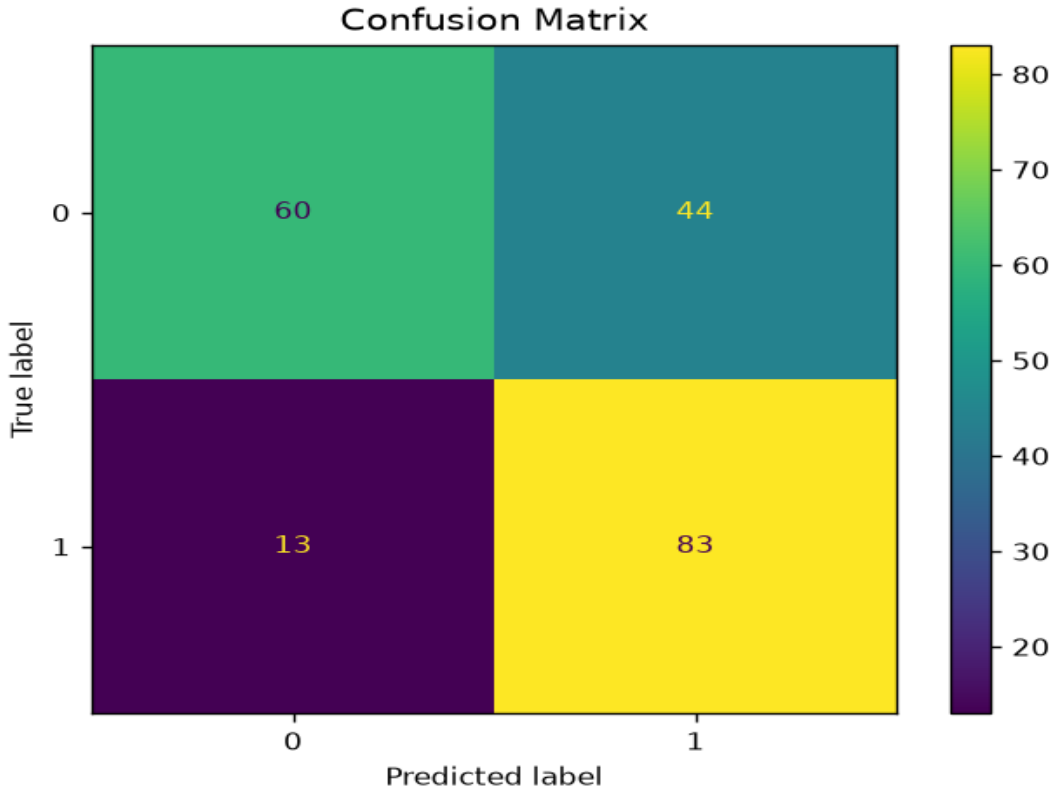
TechSupport_Yes 0.012324

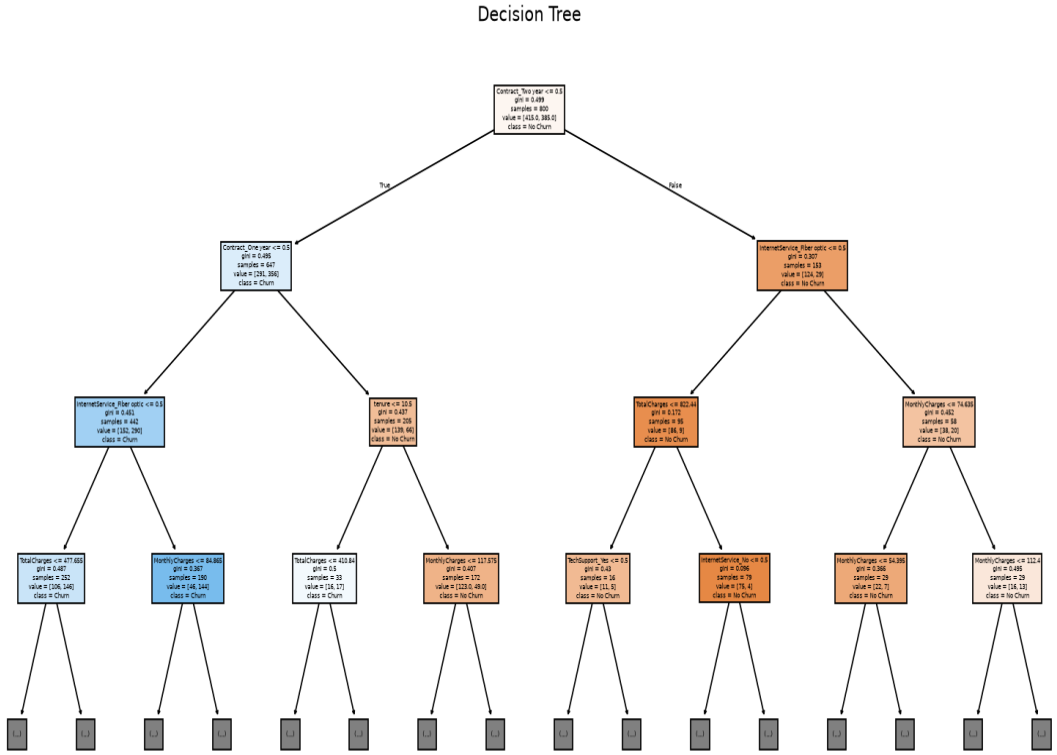
Partner_Yes 0.010306

InternetService_No 0.001825

SeniorCitizen 0.000000

dtype: float64


Confusion Matrix
60
44
13
83
0
1
True label
0
1
Predicted label
80
70
60
50
40
30
20


Decision Tree